

NOTICE NUMBER: HQ003425RFIQT

SPECIAL NOTICE

Request for Information:

Quantum Timing Technology

April 16, 2025

Issued by:

Washington Headquarters Services

Acquisition Directorate

Emerging Capabilities and Defense Prototyping

Rapid Prototype and Acquisition Branch

Table of Contents

BACKGROUND:	3
FORMAT & PAGE LIMITATION:	5
USE OF INFORMATION:	5

Request for Information (RFI) Special Notice

Quantum Timing Technology

Washington Headquarters Services Acquisition Directorate (WHS/AD)

Emerging Capabilities and Defense Prototyping (ECDP)

Responses Due: April 30, 2025, 12:00 PM (Noon, Eastern Time)

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RFI DESCRIPTION:

The Office of the Undersecretary of Defense for Research and Engineering (OUSD(R&E)) - Office of the Principal Director of Quantum Science is planning a trilateral test of quantum timing technology under the Australia-UK-US (AUKUS) partnership. Washington Headquarters Services Acquisition Directorate (WHS/AD) seeks to identify quantum timing technologies geared specifically toward defense applications and near-term platform integration.

The goal of this RFI is to identify existing quantum timing technology with the potential for platform integration that can be available for the May 2025 test in Washington, D.C.

We will investigate the environmental sensitivity of selected participating prototype clocks to provide additional information into the modelling and simulation efforts. These environmental tests will provide information necessary to plan for integration into platforms for future demonstrations of AUKUS quantum-enhanced PNT technology. These environmental tests will include operation under changes in temperature, humidity, and orientation, with the option of including a dynamic orientation test to mimic challenges of maritime environments.

We request that proposed devices be prototype devices that are not currently commercially available but that are at a Technology Readiness Level of 4 or above and have the potential to be integrated into an operational platform within the next two years. In addition to this requirement, we ask that proposed clocks meet the following technical specifications:

- $ADEV \leq 5 \times 10^{-13}$ at 1 second; $ADEV \leq 3 \times 10^{-14}$ at 10,000 seconds
- Signal output: 10 MHz
- Wall power draw $\leq 100W$
- Size $\leq 6U$

All proposed devices must be made by U.S.-based entities. The test can accommodate up to five clocks, which will be chosen based on documented performance specifications.

BACKGROUND:

OUSD(R&E) works closely with Military Services, Combatant Commands, industry, academia, and other stakeholders to ensure that the Department's science and technology strategy addresses key national security challenges. Quantum Science is a critical technology area vital to maintaining the United States national security. OUSD(R&E) focuses the Department's efforts and investments into its critical technology areas to accelerate transitioning key capabilities to the Military Services and Combatant Commands.

The goal of the Quantum-Enhanced Position, Navigation, and Timing (PNT) team within the Australian, UK, and US (AUKUS) Pillar II Quantum Working Group is to enable and accelerate robust, resilient PNT solutions through enhanced measurements and underpinning technologies, with a focus on trilateral

cooperation and interoperability. The AUKUS Clock Comparison Test, planned in April of this year at the Naval Research Lab (NRL), is an important first step towards future interoperability tests, focusing on benchmarking and comparing prototype timing technologies using standardized equipment and methodologies. We wish to offer US-based companies an opportunity to test their most promising prototype technology alongside the AUKUS devices being furnished.

Each nation under AUKUS is currently developing their own quantum timing technology. Recognizing the benefit of ensembling multiple different clock technologies together for more robust and resilient performance, the Quantum-Enhanced PNT team have proposed to harness the work of the individual AUKUS nations and focus on interoperability between clocks and time-transfer systems as a critical accelerating step.

REQUESTED INFORMATION:

Responses are welcome from all capable sources including, but not limited to, private or public companies, individuals, universities, university-affiliated research centers (UARCs), not-for-profit research institutions, and U.S. Government-sponsored labs. Respondents are encouraged to address the following to the maximum extent practicable:

- Short (1 to 2 paragraph) description of the technology proposed for consideration
- Technical specifications of technology for consideration, including size, weight, power, and necessary cables and connectors for testing
- Description of environmental test conditions the technology can endure
- Description of how the technology can be delivered to the Naval Research Lab in Washington DC, NLT May 5, 2025
- Description of how the technology will be ready for platform integration within two years and the platforms for which this technology has been designed
- Description of potential controls on the resulting data from environmental testing as it relates to sharing with both government and industry entities in the UK and Australia, as well as other US industry entities (Note: All data must be shareable with government and necessary contractors within the US, UK, and Australian government)

SUBMISSION INSTRUCTIONS

Documentation shall be delivered via email to: whs.mc-alex.ad.mbx.rpab@mail.mil, cc: Jennifer Robinson @ jennifer.e.robinson24.ctr@mail.mil and Louise Kim @louise.c.kim.ctr@mail.mil.

E-mail responses are preferred but if you have large files or information that is proprietary in nature to send, you may provide your submission using the DoD SAFE site (<https://safe.apps.mil/>). You must request a submission email for the DoD SAFE site via the listed POCs. You will then receive an email from DoD SAFE with a secure link and password to upload and submit your documents via the DoD SAFE site.

The cover material (not included in the page count) shall include the following at a minimum:

- Interested Party Name, mailing address, overnight delivery address (if different from the mailing address), phone number, and fax number;
- North American Industry Classification System (NAICS) code under which the company traditionally does business and the Business type (e.g., large business, small business) based upon that NAICS code;

- A designated point of contact of the Interested Party, including that contact's name, title, direct phone number, and e-mail address.

Responses shall be submitted electronically, in Microsoft Word (Office 2010 compatible) format or in PDF, **not later than April 30, 2025, 12:00 PM (Noon, Eastern Time)**. If late information is received, it may be considered by the Government reviewers, depending on agency time constraints.

Please note: the Government is not required to provide feedback to RFI Responders.

FORMAT & PAGE LIMITATION:

Each response to the RFI is limited to no more than 5 pages, using 12-point Times New Roman font and 1-inch margins on 8.5-inch by 11-inch page size. Each response must include a cover page (excluded from 5-page limit) with the following information:

- Interested Party Name
Mailing address
Overnight delivery address (if different from the mailing address)
Phone number
Fax number
- North American Industry Classification System (NAICS) code under which the company traditionally does business
Business size/type (e.g., large business, small business) based upon that NAICS code;
- Point of Contact Information
Name
Title
Direct Phone Number
Email Address

Responses must be unclassified and able to be shared with foreign partners from the UK and Australia. Responses containing proprietary information must be appropriately marked. The following may be included in each response and are excluded from the 10-page limitation:

- Oversize pages for graphic presentations (including foldouts)
- Existing commercial documentation
- Existing technical specifications
- Existing product literature

Note: Responses must be unclassified and able to be shared with foreign partners from the UK and Australia.. Responses may contain proprietary information but must be appropriately marked.

USE OF INFORMATION:

The purpose of this notice is to gain information for potential applications relevant to the DoD and AUKUS foreign partners. All proprietary information contained in the response shall be clearly marked. Any proprietary information contained in response to this request will be properly protected from any unauthorized disclosure. The Government will not use proprietary information submitted from any one firm to establish future capability requirements. Participants shall agree to confer and consult with the OUSD(R&E) team prior to the publication of any press release directly or indirectly related to this notice. The OUSD(R&E) team shall confer and consult with any participants prior to the publication of any press release that directly includes a written or visual reference to the name of their organization, employees, or

technology. Participants will not use the name, logo, or images of any U.S. Government organization, employee, or contractor without prior approval from the relevant organization.

Non-government advisors may be used to objectively review a particular functional area and provide comments and recommendations to the Government. All advisors shall comply with procurement integrity laws and shall sign non-disclosure and rules of conduct/conflict of interest statements. The Government shall take into consideration requirements for avoiding conflicts of interest and ensure advisors comply with safeguarding proprietary data. Submission in response to this RFI constitutes approval to release the submittal to Government support contractors.

This RFI is issued solely for information and program planning purposes. This RFI does not constitute a formal solicitation for proposals or proposal abstracts. Responses to this notice are not offers and cannot be accepted by the Government to form a binding contract. Submission of a response is strictly voluntary and does not commit the interest party to propose on any subsequent announcements or solicitations on this topic. The Government will not reimburse costs incurred for responding to this RFI. WHS and OUSD(R&E) are not obligated to acknowledge receipt or provide feedback to respondents with respect to any information submitted in response to this RFI.